

Where a Quantum Reservoir Works: A Transferable Operating Band

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Abstract—In quantum reservoir computing, a fixed quantum system transforms an input signal, while learning reduces to training a simple linear readout on its measured outputs. Since the quantum dynamics themselves are never optimized, the method is well suited to today’s hardware. Yet these dynamics must still be chosen carefully, because their settings remain fixed throughout training and inference. It therefore remains an open question where, in its control space, a fixed quantum system learns well. We address this question for a dissipative reservoir by mapping performance over three central physical controls: the strength of the input drive, the coupling between neighboring qubits, and the rate of dissipation. Good performance concentrates in a single, well-defined operating region of this control space. This region transfers across tasks and reservoir initializations, and the same memory-defined regime persists under architectural changes. It is also mechanistically grounded, since it disappears whenever any of the mechanisms that create it is removed. Finally, the region can be located cheaply before any task is run, using a simple memory diagnostic.

Index Terms—quantum machine learning, reservoir computing, quantum computing, dissipative systems, memory capacity

I. INTRODUCTION

Reservoir computing builds on a simple principle: a fixed dynamical system, left untrained, transforms an input signal into a high-dimensional record of its recent history, and learning reduces to fitting a single linear readout on top of that record [1]–[3]. The dynamics are never optimized, only observed. This simplicity is what makes the approach appealing for near-term quantum hardware: native quantum evolution can serve as the reservoir, and only the classical readout must be trained [4], [5]. At the same time, it places the full burden on the fixed dynamics, and so raises a question that remains open: what makes those dynamics useful?

Recent QRC work spans platforms from noisy gate-model devices to photonic and continuous-variable reservoirs, and protocols from weak measurement to engineered dissipation and experimental optical memory control [5]–[8], [12]. Across this range the question persists, and classical reservoir theory already supplies the warning: maximal expressivity alone does not make a reservoir useful. A reservoir works by holding a balance, retaining recent input, forgetting old input in a controlled way, and exposing nonlinear functions of that history to the readout [3], [9]. A quantum reservoir inherits this lesson but complicates it, because these ingredients are no longer free dials. They are consequences of distinct physical resources: how strongly each input perturbs the carried state, how coherently the system mixes that state across its parts, and how quickly it is allowed to forget [5], [12].

For a stateful dissipative gate-model reservoir, we treat the space of these three resources as the proper setting in which to ask where the reservoir works, and we map it. What emerges is not a single tuned operating point but a connected region, characterized by gentle input, nonmaximal coupling, and controlled damping, whose boundaries stay essentially unchanged from task to task. The contribution of this paper is that region itself: its existence, its transfer across tasks and reservoir initializations, the mechanisms that sustain it, and its predictability, since a simple memory diagnostic locates it before any task-specific search.

II. PROTOCOL

A. Reservoir

The reservoir is a small qubit system run as a recurrent loop. At each step it receives one value of a scalar input series, evolves, and hands its state forward without ever being reset, so the carried state holds a record of recent inputs. We write $\rho_{t,\ell}$ for the state after input t has passed through layer ℓ ; with L the final layer, $\rho_{t-1,L}$ is the state handed forward from the previous step.

At step t , the input u_t is written into this carried state by the same Y rotation on every qubit,

$$\rho_{t,0} = U_{\text{in}}(u_t) \rho_{t-1,L} U_{\text{in}}^\dagger(u_t), \quad U_{\text{in}}(u_t) = \bigotimes_i R_y(\beta u_t), \quad (1)$$

where the input drive β sets how strongly each step writes new input into the carried state.

The perturbed state then passes through L identical layers, each a frozen local mixer U_{mix} , a ring ZZ coupling U_λ , and amplitude damping D_γ applied to every qubit,

$$\rho_{t,\ell} = D_\gamma^{\otimes n} \left[U_\lambda U_{\text{mix}} \rho_{t,\ell-1} U_{\text{mix}}^\dagger U_\lambda^\dagger \right], \quad U_\lambda = \prod_{(i,j) \in E_{\text{ring}}} e^{-i\lambda Z_i Z_j}, \quad (2)$$

with E_{ring} the nearest-neighbor edges of the qubit ring, λ the coupling strength, and γ the damping rate. Everything else is frozen: we vary only $\theta = (\beta, \lambda, \gamma)$, with β and λ angles in radians and $\gamma \in [0, 1]$ the amplitude-damping probability, and train only a linear readout. Each control supplies one ingredient reservoir theory requires: β sets how strongly each new input is written in without overwriting what the state already carries, λ generates nonlinear mixing of past inputs, and γ provides the controlled forgetting on which fading memory depends.

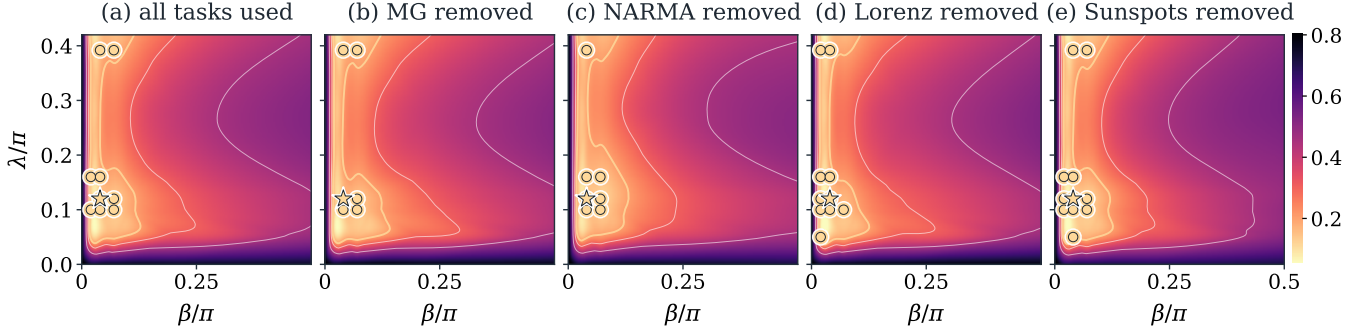


Fig. 1. The good operating region is essentially the same whichever task we score. Each panel maps validation rank over input drive β and coupling λ at fixed damping $\gamma = 0.12$. Brighter means better rank. Panel (a) uses all four tasks; panels (b) to (e) each drop one. Gold circles mark the top validation quintile across tasks and seeds; the star is the band’s medoid. The basin and circles barely move when a task is dropped, so the band is not an artifact of any single task.

B. Configuration

Unless stated otherwise, the reservoir has four qubits and two layers with ring coupling, uniform input injection, and amplitude damping; the readout is ridge regression on single-qubit Z and ring-pair ZZ expectations alone. We sweep the three controls β , λ , γ on a fixed grid and select every hyperparameter, including the ridge penalty, on validation data only; the holdout never defines the operating region. Crossing the four benchmark tasks described below with twenty frozen-reservoir seeds yields a balanced panel of 80 task-seed replicates per grid coordinate. All grid coordinates and task-seed replicates are fixed before performance is inspected. Transfer is assessed by leave-one-out tests over tasks and seeds, where the operating region is reconstructed without the held-out task or seed and evaluated only on what was withheld.

C. Tasks, Metric, and Splits

We use four scalar one-step-ahead prediction tasks spanning distinct demands: the Mackey–Glass and Lorenz systems for smooth chaos [13], [14], NARMA for nonlinear memory [15], and annual sunspot counts for irregular real-world data. Inputs are scaled to $[-1, 1]$ and targets standardized; each task is split chronologically into washout, train, validation, and holdout segments. We score with normalized mean-squared error (NMSE), the mean squared error divided by the target variance, so that predicting the mean is the no-skill baseline and lower is better. NMSE is computed separately on validation and holdout, and the ridge penalty is selected by validation NMSE alone.

D. Operating Band

The four tasks produce errors on very different scales, so raw NMSE values cannot be pooled across them; following standard practice for comparisons across data sets [10], we compare settings by rank instead. Within each task j and seed s , a setting $\theta \in \Theta_{\text{grid}}$ receives the validation percentile rank $r_{j,s}(\theta)$, and lower is better. A second choice guards against flukes: the single best-ranked coordinate may owe its win to one easy task or one lucky seed, so rather than crowning a winner

we keep every setting that ranks highly *often*. The operating band contains the settings that land in the top fraction p for at least a fraction q of task-seed pairs,

$$B_{p,q} = \{\theta : \Pr_{j,s}[r_{j,s}(\theta) \leq p] \geq q\}. \quad (3)$$

We call $B_{p,q}$ the band, and the dynamical behavior it identifies the operating regime. This frequency criterion is the same idea as stability selection [11], which keeps the variables chosen often across resamples; here it treats the band as a search prior over settings that reliably work. We report the strict band $B_{20,0.7}$, the settings that reach top-quintile rank on at least 70% of the task-seed replicates. We chose these thresholds by convention before running the analysis, and the band is not sensitive to them: relaxing either one enlarges the same region rather than revealing new ones, from 22 grid coordinates at $B_{20,0.7}$ to 41 at $B_{20,0.6}$ and 70 at $B_{30,0.7}$. In the transfer tests the band is built from the remaining replicates and scored on the withheld one; in the holdout audit it is frozen on validation, then evaluated once on holdout. Because ranks are formed within each task-seed replicate before aggregation, no large-error task or difficult seed can dominate the band.

III. RESULTS

Across the control space, low validation rank gathers into one compact, connected basin rather than scattered optima, and that basin barely moves when any single task is dropped (Fig. 1). The damping coordinate is selected by validation: the near-zero slice $\gamma = 0.01$ contains almost no band points, while the strict $B_{20,0.7}$ core fills in from $\gamma = 0.12$ to 0.30 with medoid $\gamma = 0.22$ (Fig. 2). We plot $\gamma = 0.12$ in Fig. 1 because it is the first damping value at which the band is populated.

The band built on these data survives their removal. To test whether it generalizes beyond a joint fit to the four tasks, we rebuild it with part of the data withheld and score it on exactly that part. When one task is withheld, the band built from the rest scores mean held-out rank 0.193 with CI [0.176, 0.210]; when one seed is withheld, the corresponding mean rank is 0.151 with CI [0.139, 0.163]. Since rank is normalized so that 0.5 is chance, both values sit deep in the top quartile: the

TABLE I

HOLDOUT PERFORMANCE AFTER VALIDATION-ONLY BAND SELECTION. RANK IS NORMALIZED SO THAT 0.5 IS CHANCE AND LOWER IS BETTER.

Task	Band NMSE	Medoid NMSE	Band rank
Mackey–Glass	7.05e-05	5.26e-05	0.104
Lorenz	1.51e-05	1.52e-05	0.136
NARMA	0.560	0.607	0.204
Sunspots	0.209	0.209	0.168
All	0.192	0.204	0.153

band transfers across task identity and reservoir initialization alike. It also holds on unseen data: frozen on validation and evaluated once on the chronological holdout, it remains in the top quartile on every task (Table I). The regime even persists beyond a single architecture. A four-qubit, three-layer variant forms a band whose held-out performance matches the base model (0.154 vs. 0.153) even though its exact grid coordinates shift (Jaccard overlap 0.04): what carries over is the regime, the balance of drive, coupling, and damping, not a fixed list of coordinates. To test whether the band reflects the reservoir mechanisms rather than a coincidental grid location, we repeat the selection after ablating one ingredient at a time: removing damping, removing the recurrent mixer, replacing amplitude damping with dephasing or depolarizing noise, or restricting the readout to Z -only or ZZ -only features. Figure 3 visualizes three representative cases: no damping, no mixer, and dephasing. The remaining ablations are run under the same criterion but are not plotted for space. In the base model, the plotted $\gamma = 0.12$ slice contains nine band coordinates with mean rank 0.223, well inside the top quartile. Under every ablation, however, the re-selected band is empty: no coordinate is chosen often enough across task–seed replicates to satisfy the same $B_{20,0.7}$ stability criterion. Thus, none of the ablated reservoirs retains a reproducible operating band under the original selection rule.

The original band coordinates also lose their advantage once the mechanisms are removed. Their mean rank deteriorates to 0.473 without damping, 0.445 with dephasing, 0.509 with depolarizing noise, 0.367 without the recurrent mixer, 0.486 with Z -only readout features, and 0.468 with ZZ -only features. Even the mildest case falls outside the top quartile, and most sit near chance. This rank loss concentrates exactly where the base band sat (Fig. 3), showing that the ablations break the region that was useful in the original reservoir. Taken together, the results indicate that the operating regime emerges from the joint action of write-without-overwrite input, nonlinear mixing, controlled forgetting, and a readout that can access both local and pairwise observables.

Finally, the regime can be located before any task is run. The task-free memory map reproduces the band from the reservoir’s dynamics alone (Fig. 4a), and across all settings and seeds memory capacity is a strong predictor of validation rank ($\rho_s = -0.88$), as are information-processing capacities [9], whereas raw feature diversity is not (Fig. 4b,c). A cheap memory measurement therefore suffices to screen for good settings in advance of any task-specific search.

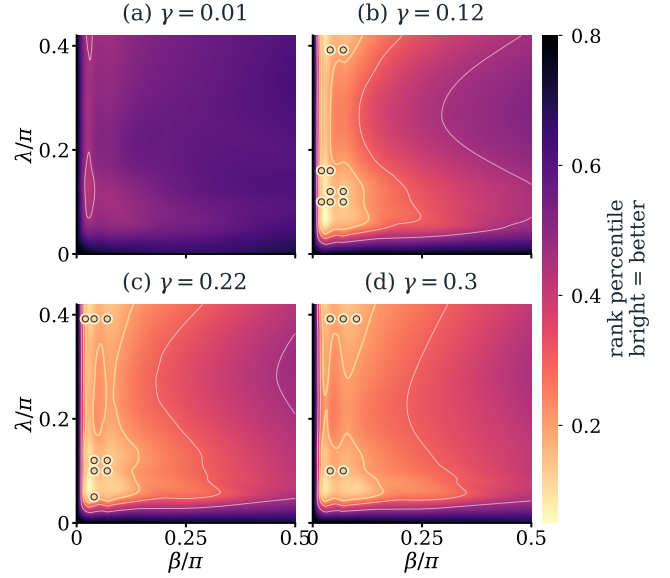


Fig. 2. Controlled forgetting is necessary for the regime. Each panel maps validation rank over (β, λ) at the damping rate γ named in its title (brighter = better); gold circles are the $B_{20,0.7}$ band points at that γ . The band is essentially empty at $\gamma = 0.01$ and fills in from $\gamma = 0.12$ to 0.30.

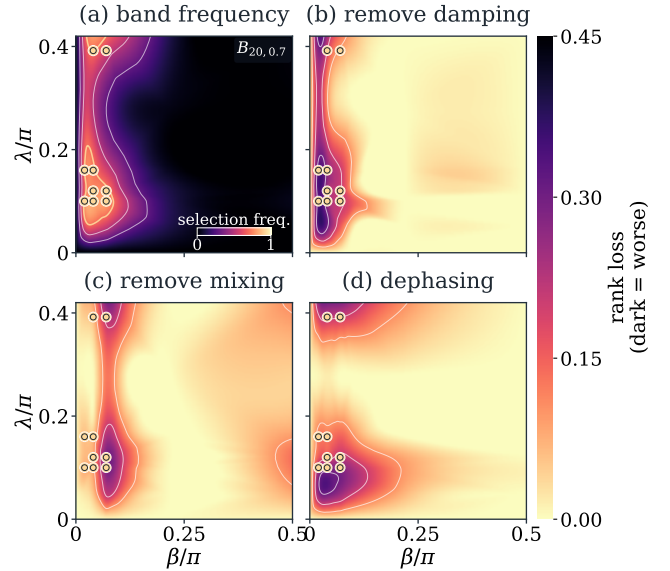


Fig. 3. Mechanism ablations destroy the operating band. (a) Selection frequency in the original reservoir: for each coordinate (β, λ) , the color shows how often it reaches top-quintile validation rank across the 80 task–seed replicates; gold circles mark the $B_{20,0.7}$ band. (b)–(d) Validation-rank loss for three representative ablations named in the panels, with darker colors indicating larger deterioration. The strongest losses occur at the original band coordinates.

IV. DISCUSSION

For this reservoir family, where the reservoir works is set by a transferable operating regime, not by task-specific tuning: four tasks spanning smooth chaos, nonlinear memory, and irregular real-world data obey the same region, and that region holds when any one task or seed is withheld. The regime is picked out by memory; neither Hilbert-space size nor raw feature diversity

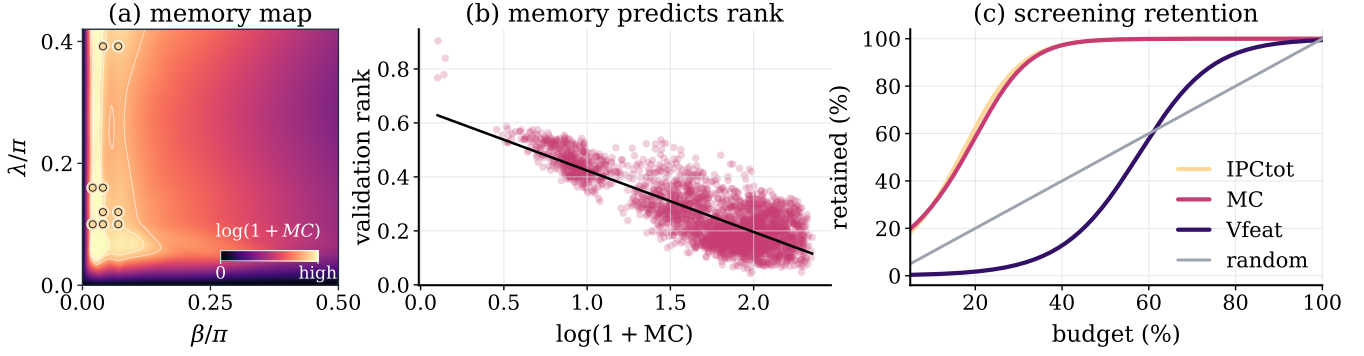


Fig. 4. Memory alone locates the band before any task is run. (a) Task-free memory-capacity map, $\log(1 + MC)$, on the $\gamma = 0.12$ slice of Fig. 1; gold points lie in the memory-rich region. (b) Higher memory predicts lower validation rank (Spearman $\rho_s = -0.88$). (c) Screening by each diagnostic’s top-ranked fraction: MC and information-processing capacity (IPC) recover good settings far faster than random selection, whereas raw feature diversity (V_{feat}) barely beats chance.

predicts it. What matters is the balance classical reservoir theory identified, with drive, coupling, and dissipation holding memory and nonlinearity in tension [3], [9]. The same balance explains the rest: strip out any single mechanism and the regime collapses, while a task-free memory diagnostic locates it before any task is run.

Our claims are restricted to simulated instances of this architecture family, with measurement assumed noiseless and freely available; finite-shot effects, calibration drift, device noise, and readout cost are not addressed here [7]. Within these limits, the practical consequence is concrete. The default workflow in QRC, a fresh hyperparameter search for every task, treats each problem as if the reservoir had to be rediscovered from scratch. Our results say it does not: one memory measurement, taken once and without any task data, identifies the settings that then serve every task we tested. How useful a dissipative quantum reservoir is depends far more on where it is operated than on how large it is, and where to operate it can be read off its memory alone.

REPRODUCIBILITY APPENDIX

A. Package

The anonymous reproducibility repository contains the material needed to reproduce every number and figure in this paper: the simulator, selection and diagnostic code, fixed grids and seeds, checked-in CSV/JSON results, and figure scripts, with all stochastic components seeded for deterministic reruns.

B. Simulator and diagnostics

We simulate the full density matrix exactly, without finite-shot sampling. The state starts from $\rho_0 = |0 \cdots 0\rangle\langle 0 \cdots 0|$, is carried between input steps without reset, and each seed uses one fixed local mixer $U_{\text{mix}} = \bigotimes_i R_x(\theta_i)$ with $\theta_i \sim \mathcal{U}[0, 2\pi)$, reused at every step and layer. The readout uses exact Z_i and nearest-neighbor ring $Z_i Z_j$ expectations plus an unpenalized intercept. All model selection, including ridge penalties and band construction, uses validation NMSE only; holdout is evaluated only after the operating band is frozen. The strict band

$B_{20,0.7}$ contains 22 coordinates in the full three-dimensional (β, λ, γ) grid, while the mechanism-ablation maps in Fig. 3 use the plotted $\gamma = 0.12$ slice containing 9 of them. Task-free memory and information-processing diagnostics are computed on an independent random drive with chronological splits; negative held-out R^2 values are clipped to zero.

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